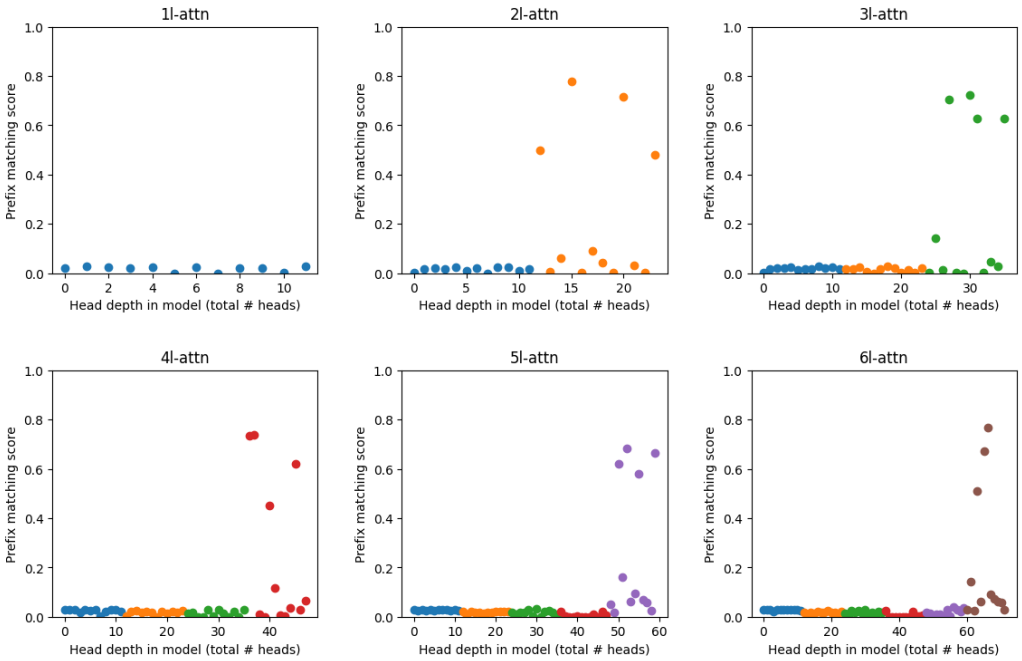
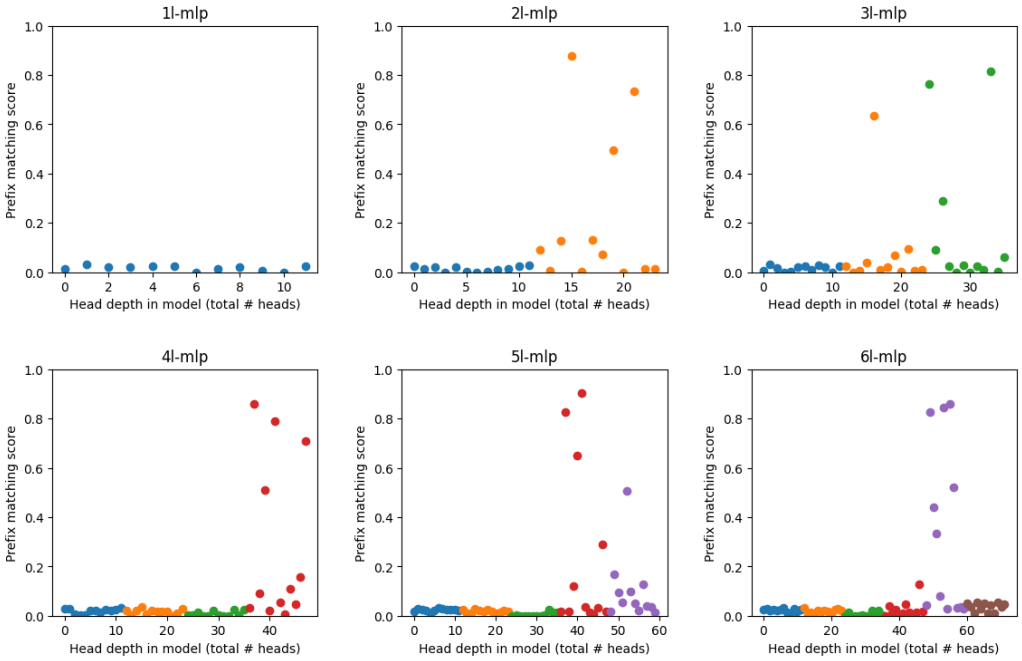


Small models with MLPs:

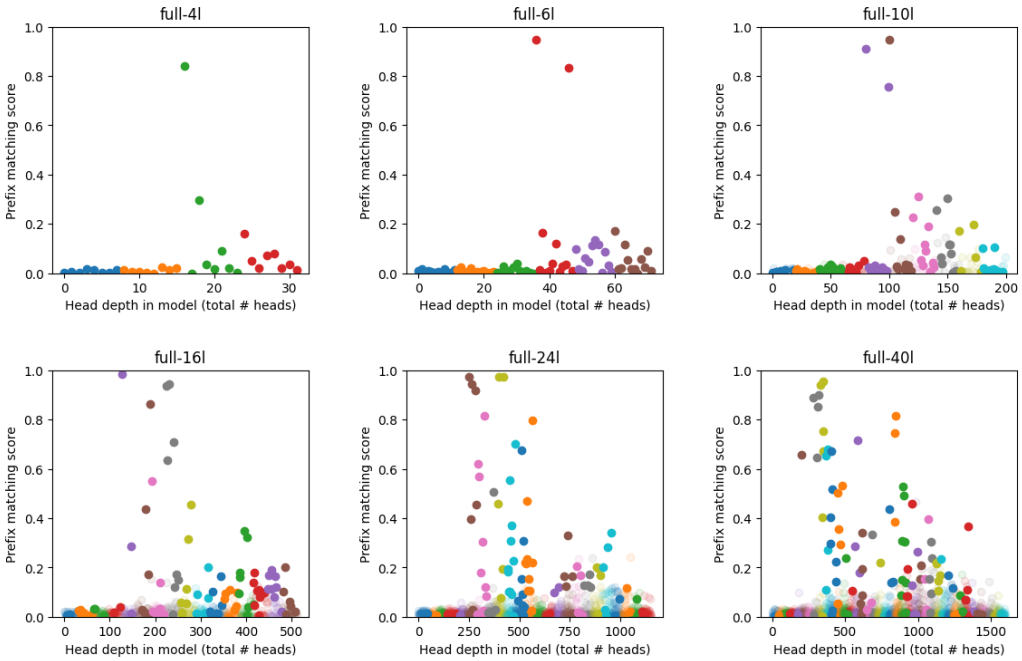
In small models with MLPs, by the 5L and 6L size, the induction heads form more in the *second-to-last* layer.



Full-scale models:

In the models in our full-scale sweep, induction heads form earlier. In fact, in 24L and 40L models the majority of induction heads form before the halfway point in model depth.

Note that in these plots, only a randomized sample of 100 heads (with selection biased towards those scoring highly on prefix matching) are shown as opaque dots; most of the datapoints are left translucent for ease of reading.

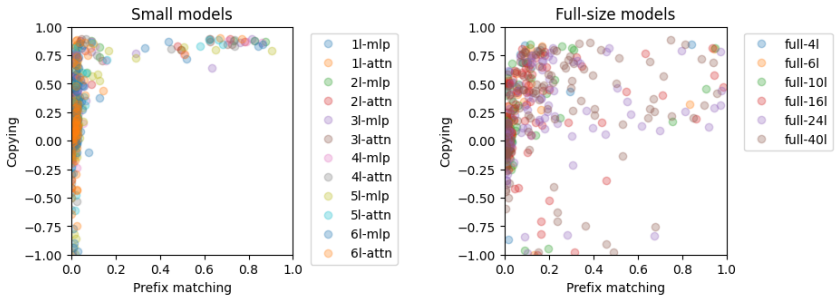


Distribution of scores

What is the distribution of heads' scores on our head activation evaluators?

The distribution is sharper in our small models than our full-size models. In both cases, heads with a positive prefix matching score are more likely to have a positive copying score, and this correlation is even stronger for the highest prefix matching scores.

(For models with more than 100 heads, a subset of heads is plotted, as seen in the opaque versus translucent datapoints immediately above.)

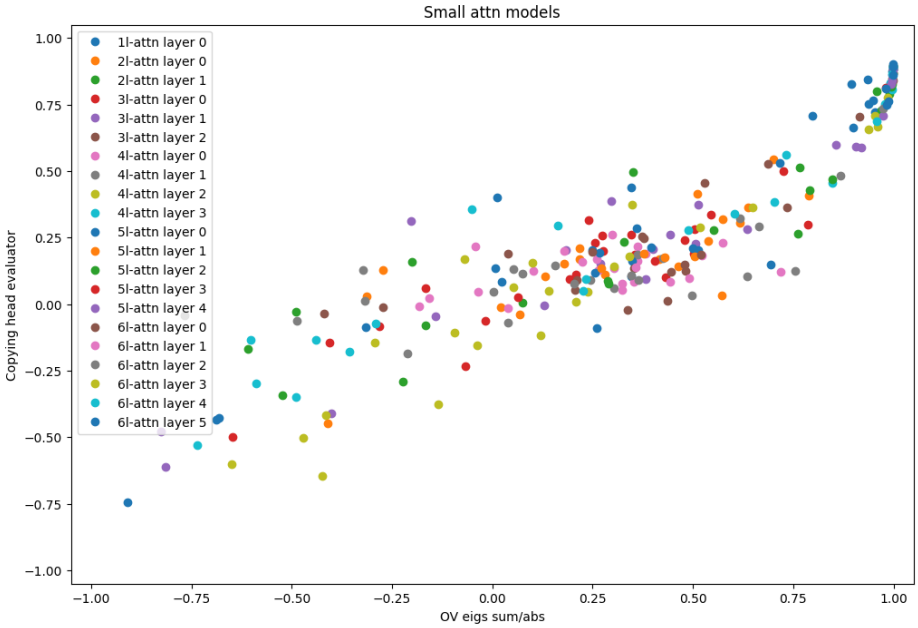


Validating head activation evaluators

We show here that our head activation evaluators (that we use to measure the properties in our definition of induction heads) correlate well with mathematically-based measures in small attention-only models.

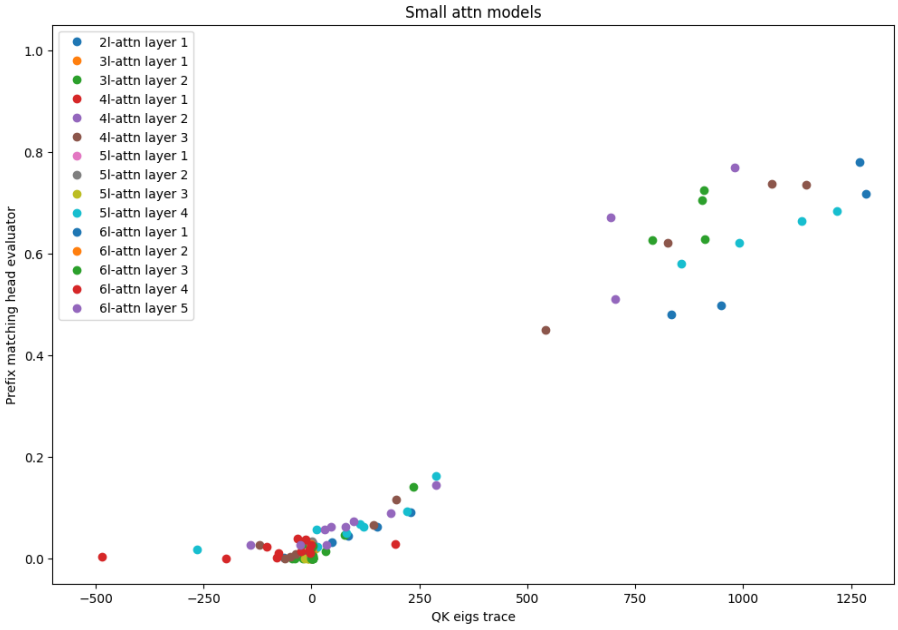
Copying

Our copying evaluator correlates well with the OV matrix of each head. Specifically here we plot against the sum of the eigenvalues, divided by the absolute value of the eigenvalues, which is a measure of what fraction of the eigenvalues are positive vs. negative.



Prefix matching

Our prefix matching evaluator correlates well with the trace of QK eigenvalues for the previous-token QK-circuit term.



Data Collection

We measure several properties of the model's overall behavior and of each constituent attention head. With the exception of the attention head ablations (which are the most computationally-intensive, and so were run only on the small models), we measure these properties on all snapshots of all models across training (that is, 200 snapshots each for the twelve small models, and 14 or 15 snapshots each for the six full-scale models). By measuring them on each snapshot, we can see how each property evolves over the course of training, rather than simply measuring the final model.